SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 1 – Interactive Tree Simulator
Weightage:	30% of CIA	**Total Marks:**	100
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.*		
Student Name:	SasikumarMegha	**Reg. No.:**	192471038
Section / Batch:	3rd cs&bs	**Date:**	29/07/2026

Code of Conduct

I ________________Sasikumar Megha____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ___Sasikumar Megha_________

Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	1	20
Tree Traversals	Traversal types, In order, Post order and Pre order	2	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	3	20
B Tree, 2-3 tree, 2-3-4 tree	Properties, structures and Operations	4	20
Red Black Tree	Properties, Example and Operations	5	20
	GRAND TOTAL:	**100 Marks**	

Annexure A – Interactive Tree Simulator

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1. Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80
   Perform:
   a. Inorder Traversal
   b. Search for 60
   c. Delete 30
   d. Display the final BST.
2. Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals. a. Inorder
   b. Preorder
   c. Postorder
3. Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.
4. Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75
5. Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

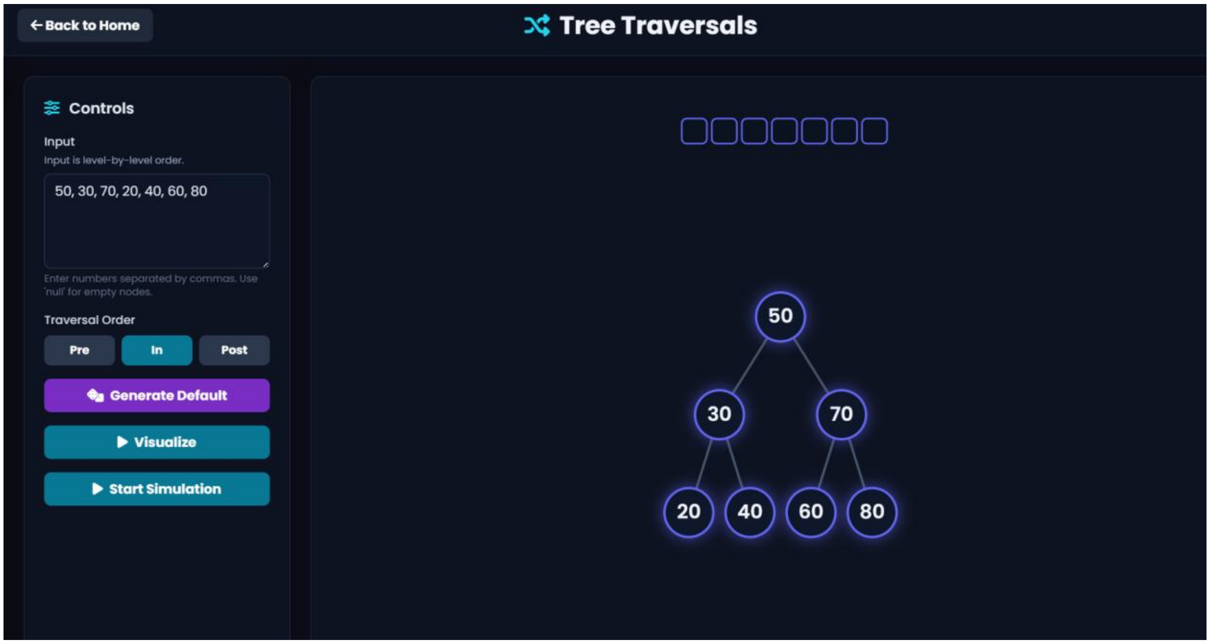
Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct Tree Construction	20	Tree is constructed correctly with all operations performed accurately	Minor errors in construction	Some operations incorrect	Tree construction mostly incorrect
Understanding of Tree Operations	20	Clearly explains insertion, deletion, search and balancing operations	Explains most operations correctly	Limited understanding	Unable to explain operations

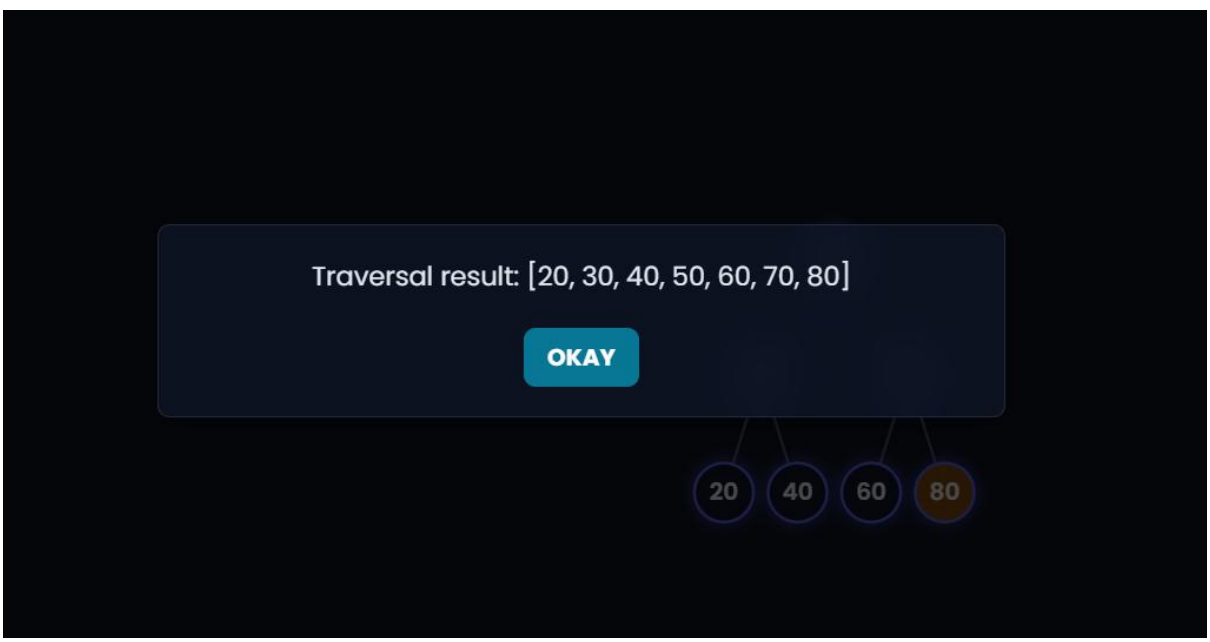
Analysis of Tree Behaviour	30	Correctly analyses balancing, rotations, node split/merge, splaying	Good analysis with minor omissions	Partial analysis	Poor or incorrect analysis
Presentation	20	Well-organized report with accurate observations and conclusion	Good report with minor issues	Basic report with limited observations	Incomplete report
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

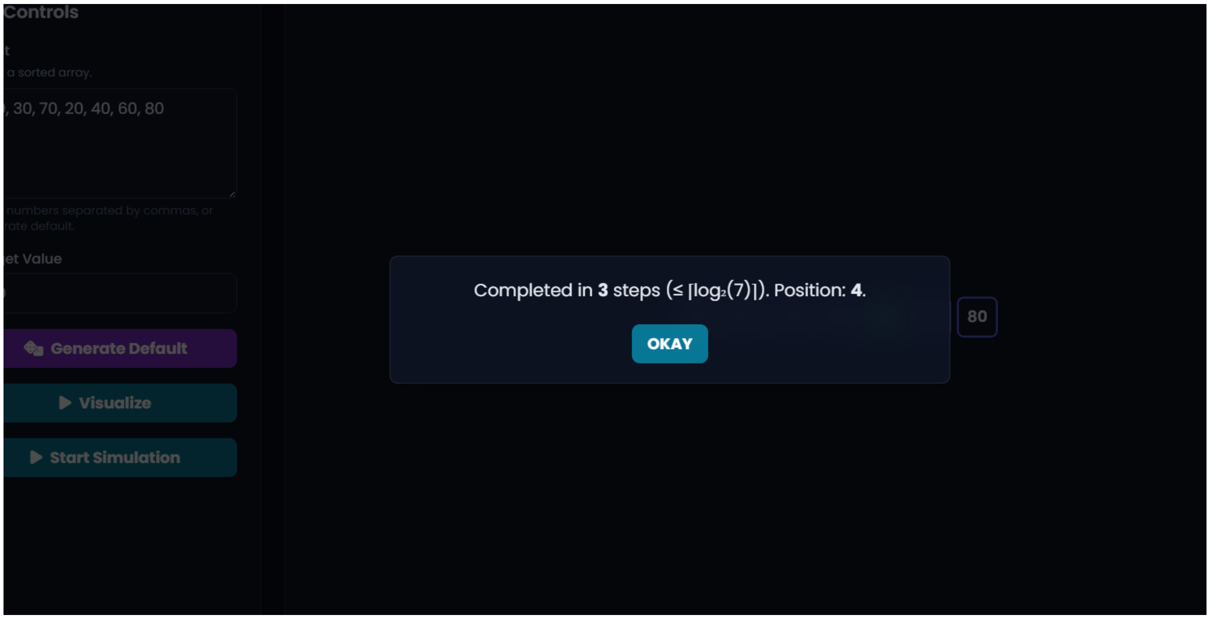
1.Binary Search Tree (BST)



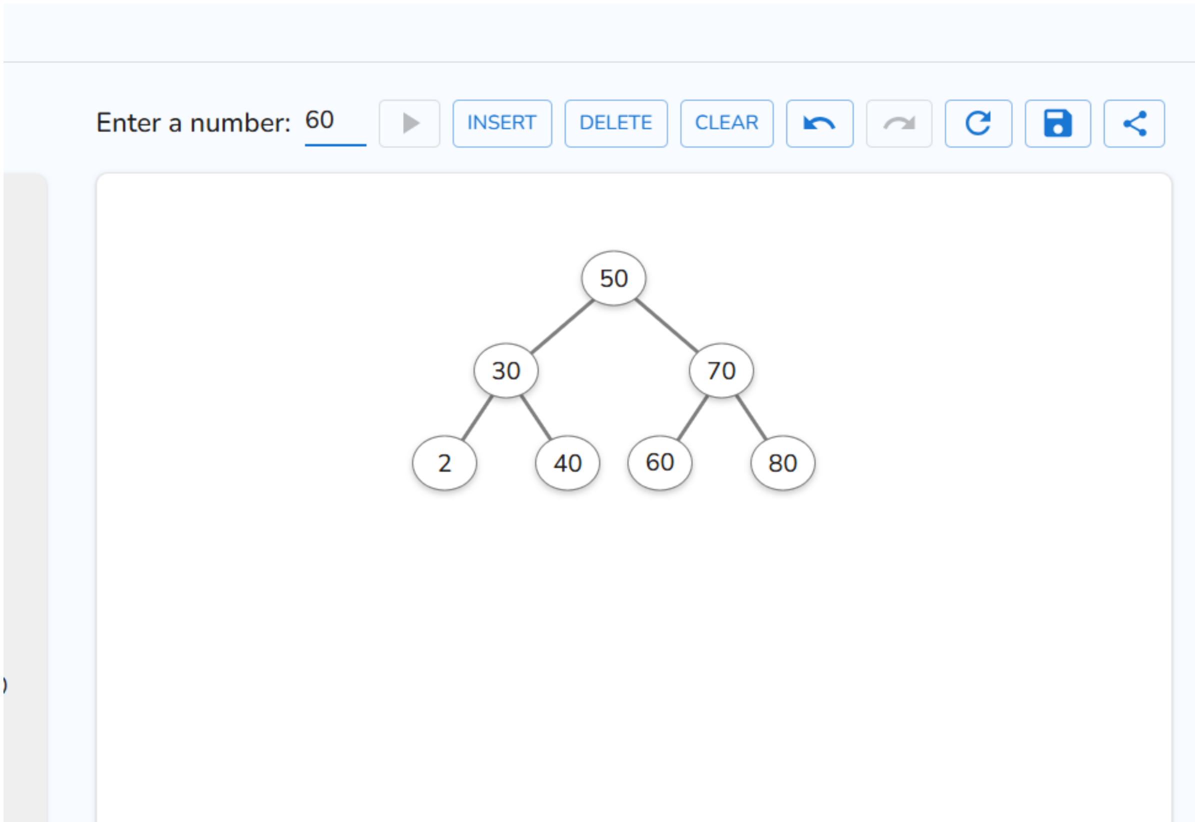
a)Inorder Traversal

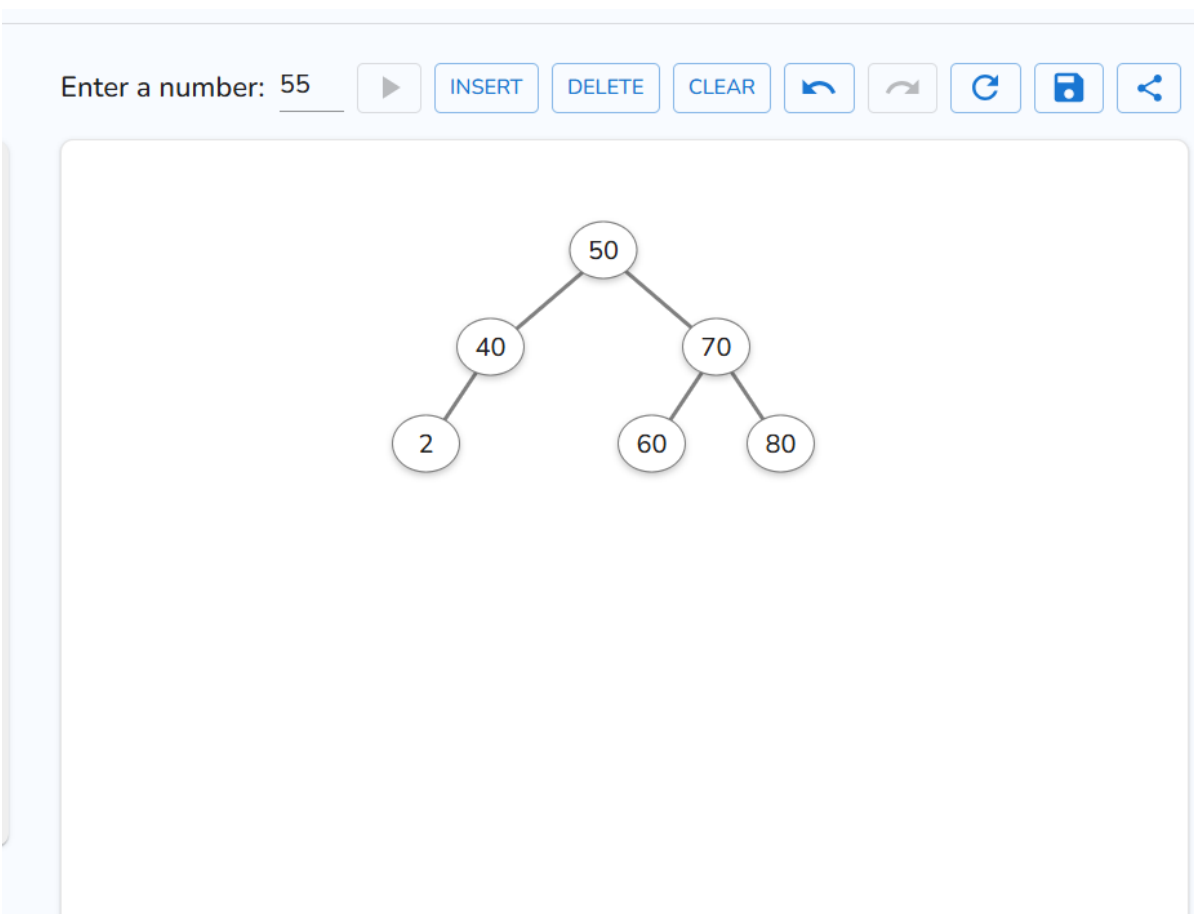


b)Search for 60

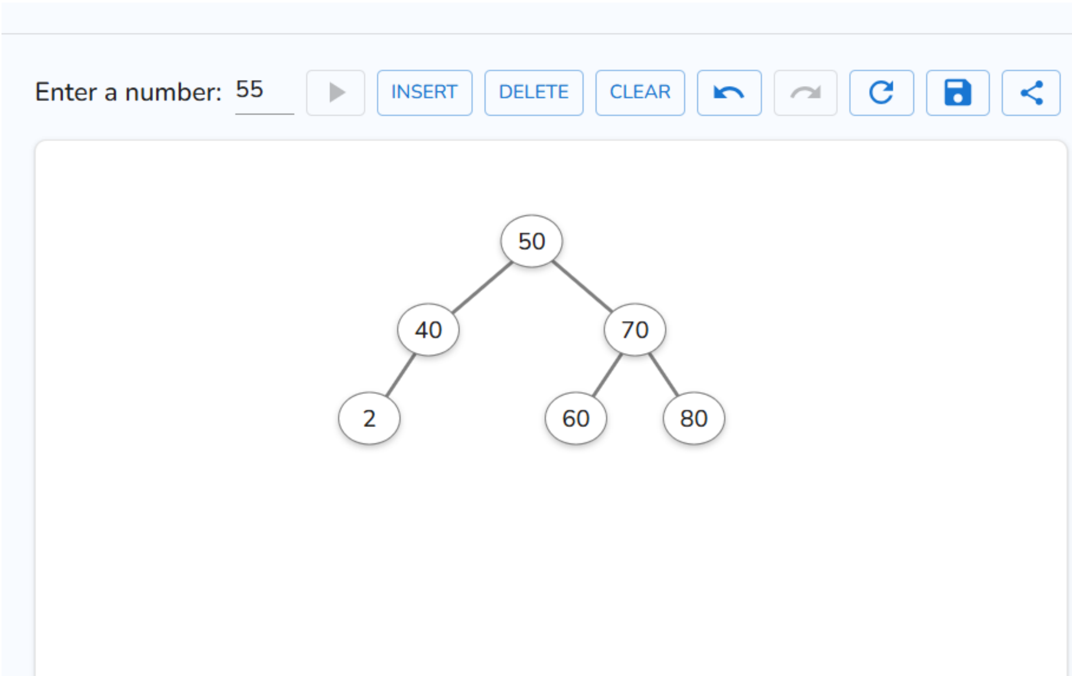


c)Delete 30


Enter a number: 60
INSERT
DELETE
CLEAR
50
30
70
2
40
60
80

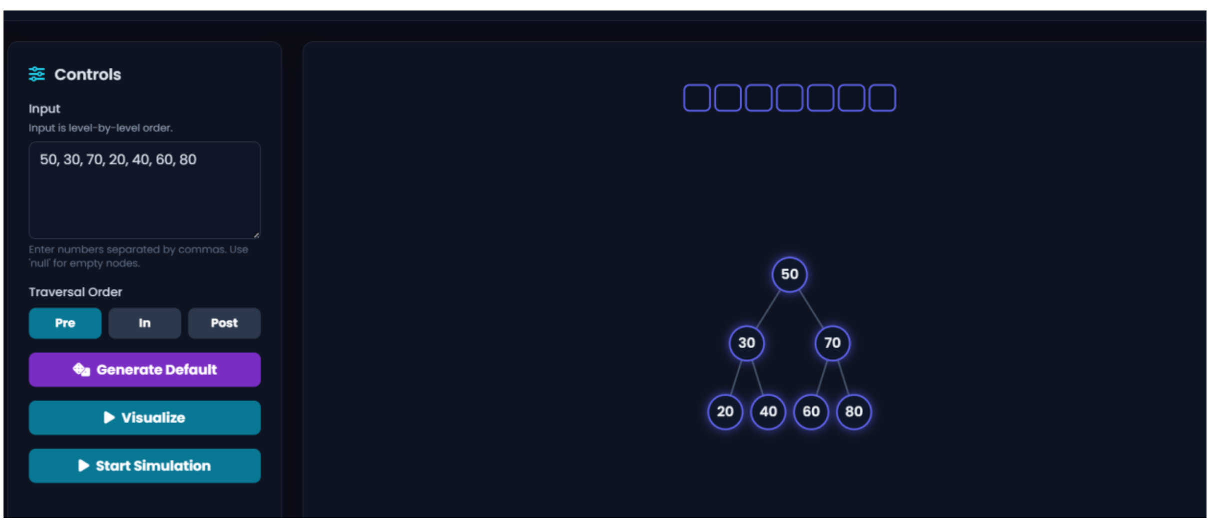

Enter a number: 55
INSERT
DELETE
CLEAR
50
40
70
2
60
80

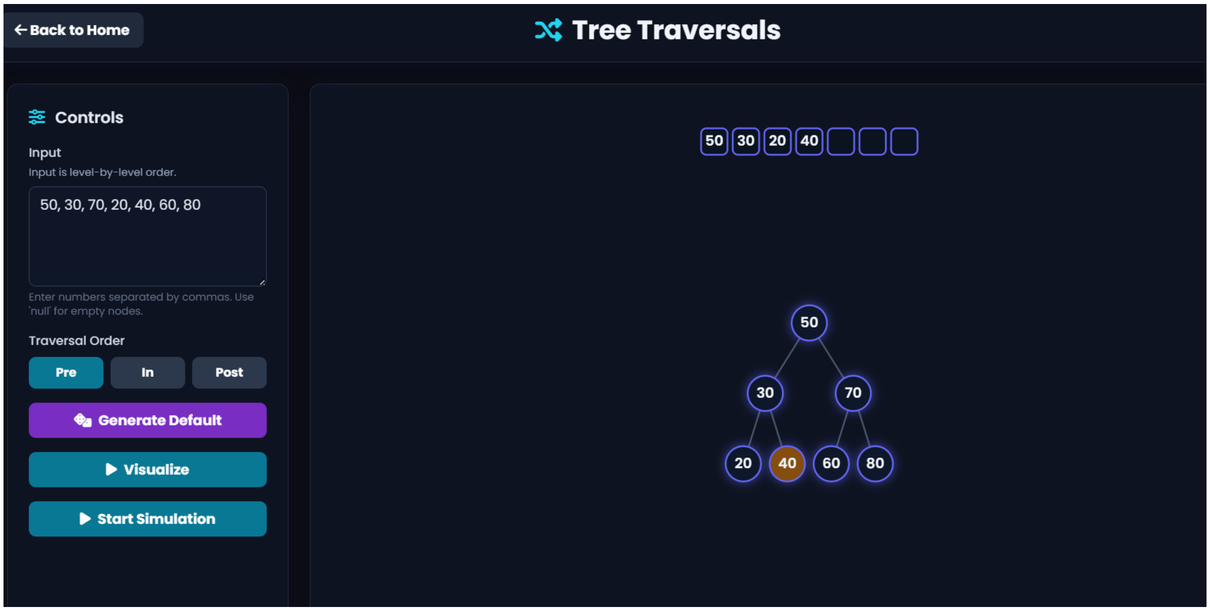
D)Final BST

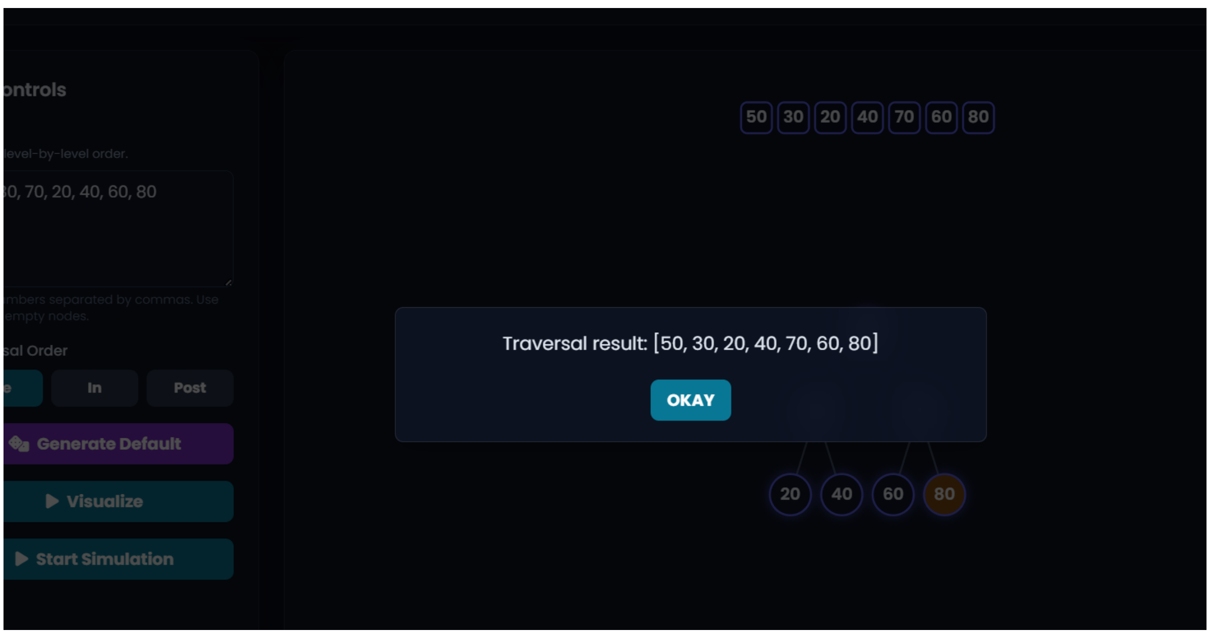


2. Complete Tree Traversals

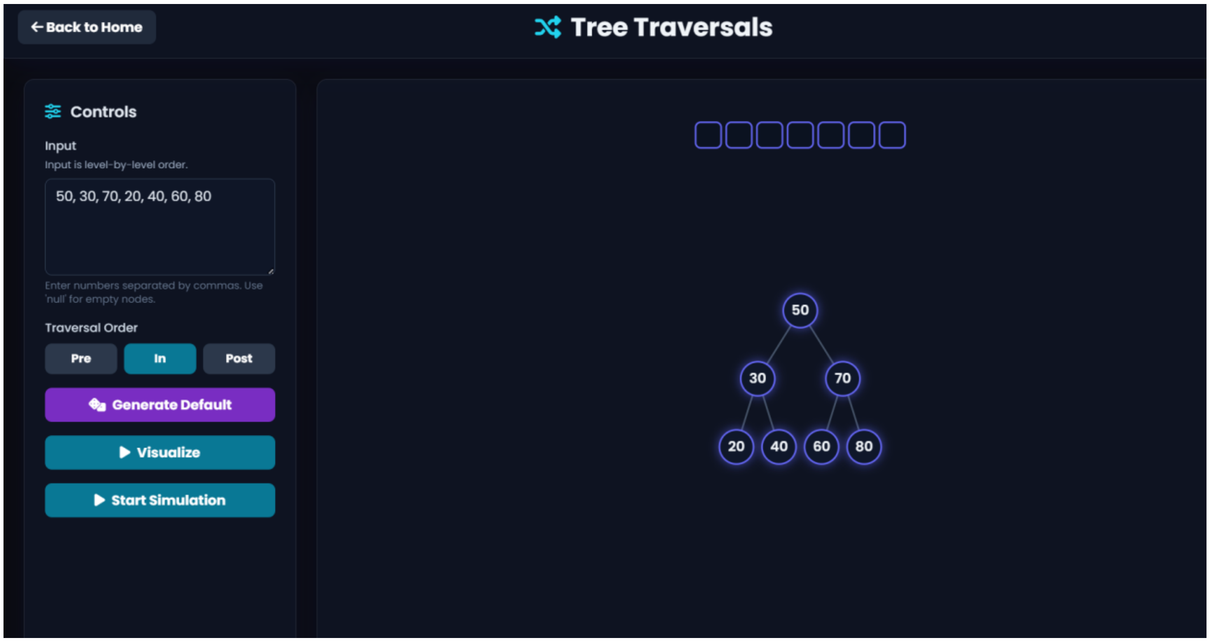
PREORDER



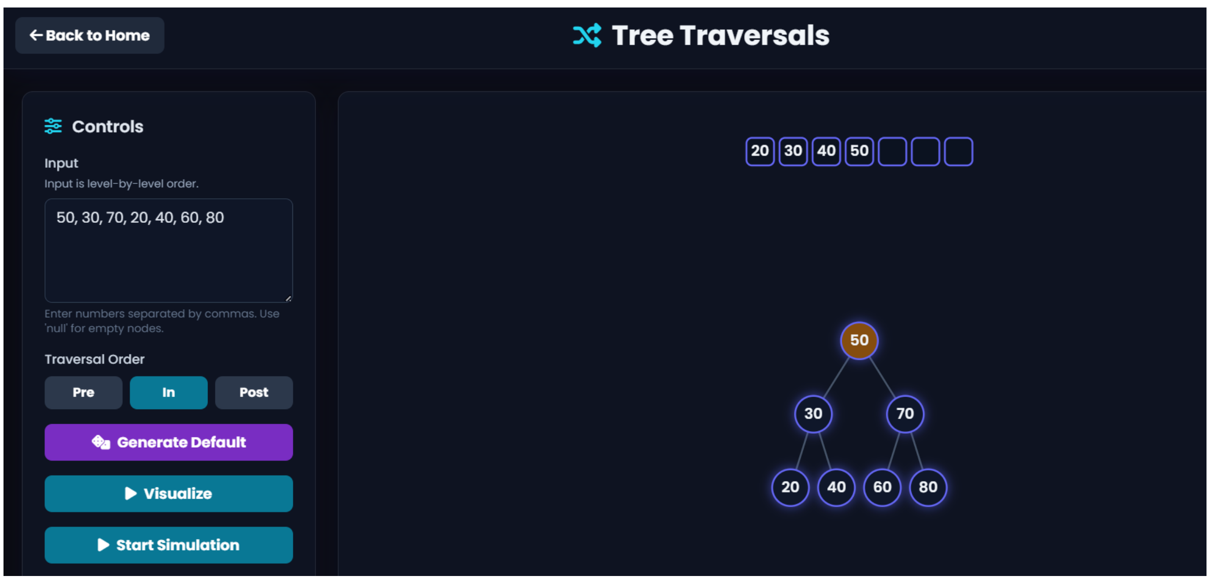

Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
50 30 20 40
50
30
70
20
40
60
80


50 30 20 40 70 60 80
Traversal result: [50, 30, 20, 40, 70, 60, 80]
OKAY
20
40
60
80

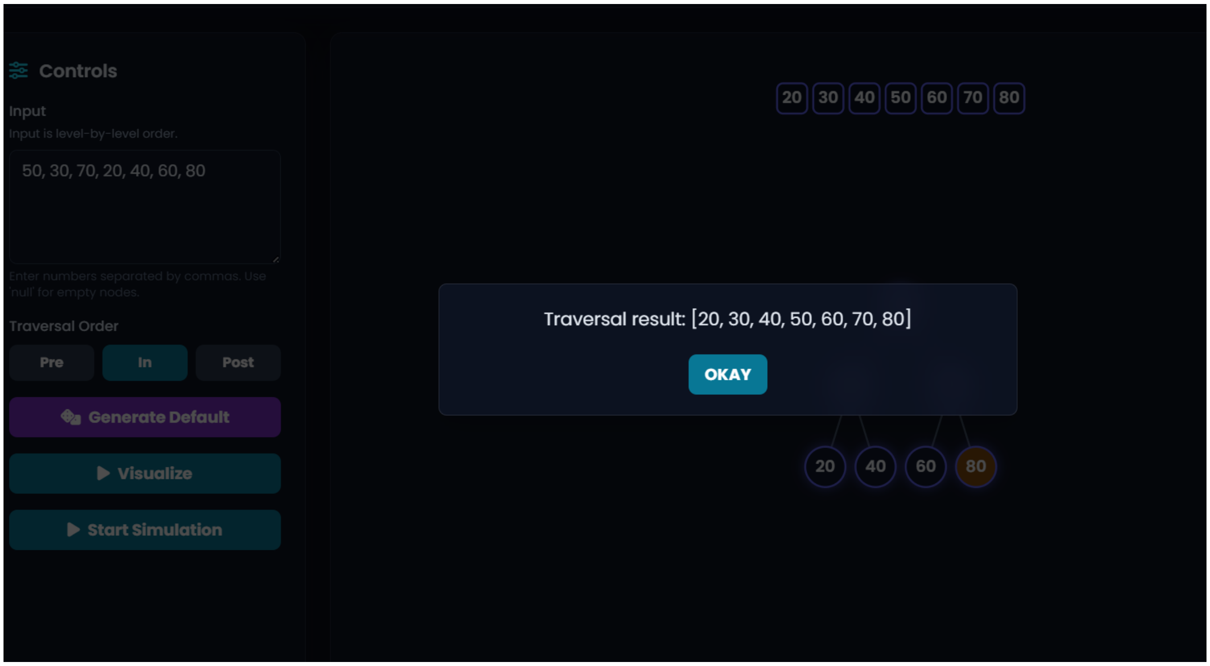
a)INORDER



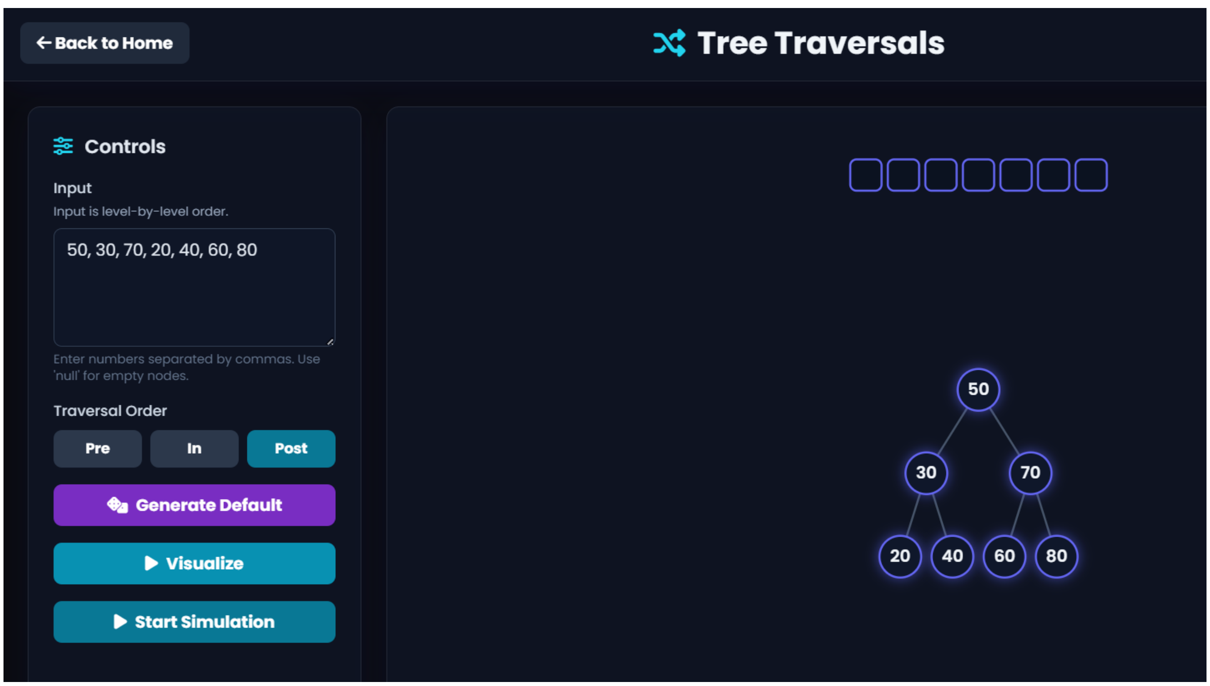
Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
50
30
70
20
40
60
80

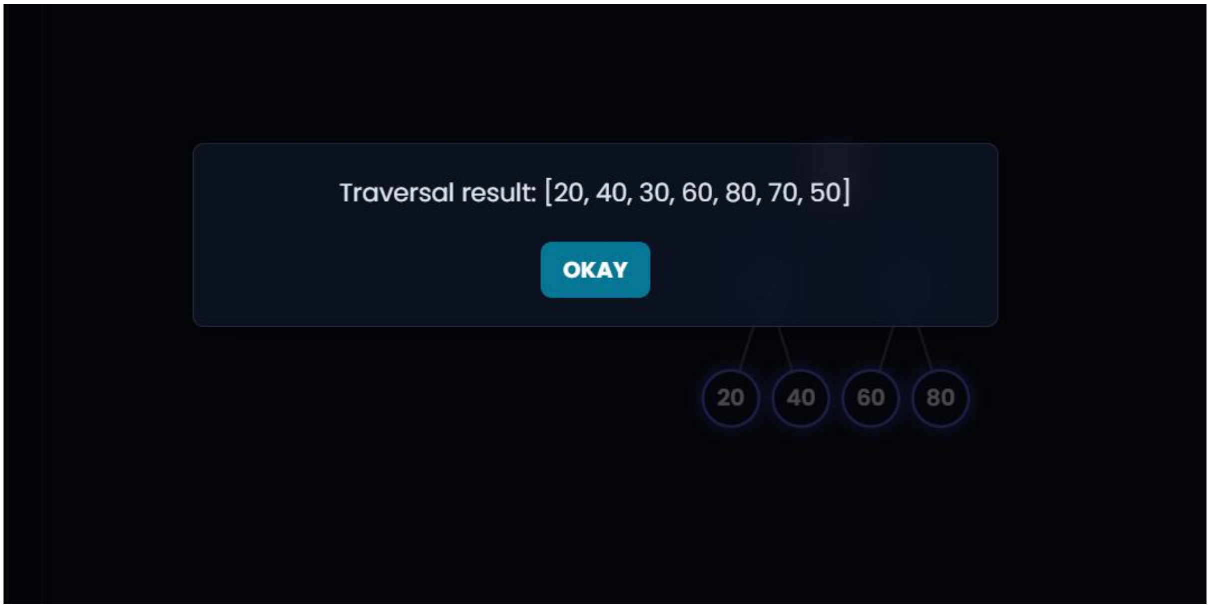


Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
20 30 40 50
50
30
70
20
40
60
80

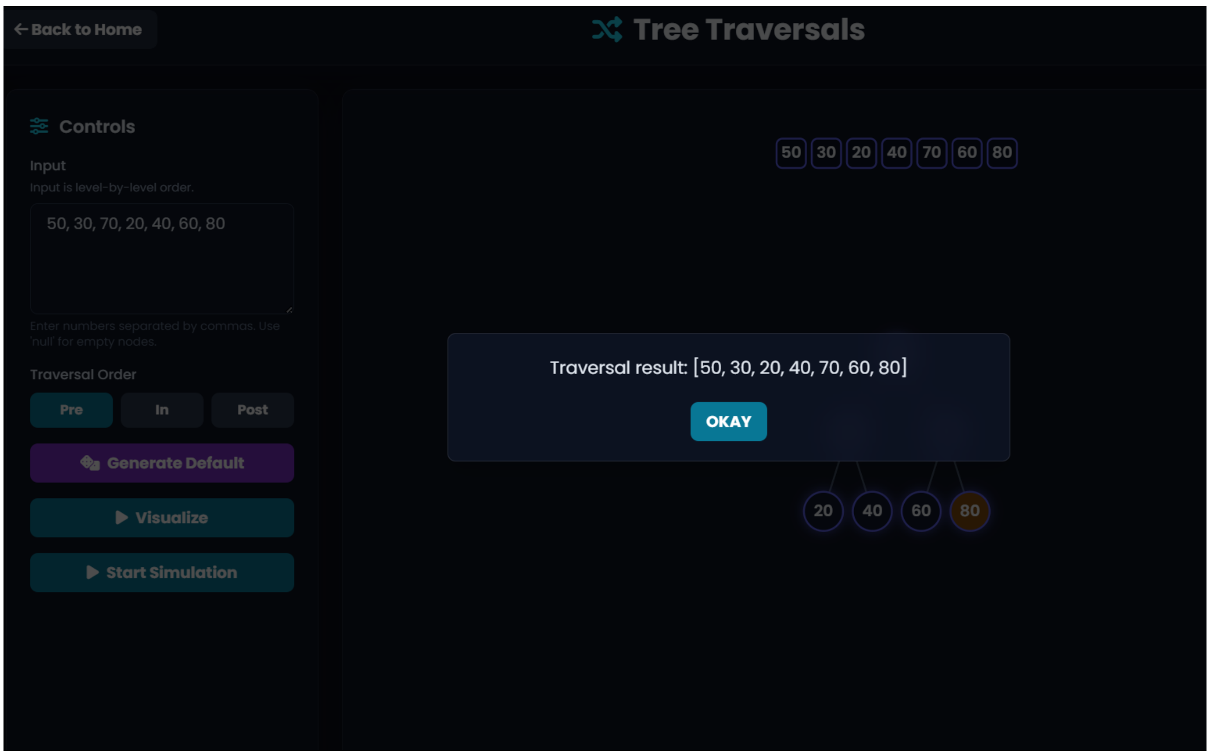

Controls
Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
20 30 40 50 60 70 80
Traversal result: [20, 30, 40, 50, 60, 70, 80]
OKAY
20 40 60 80

b)POSTORDER

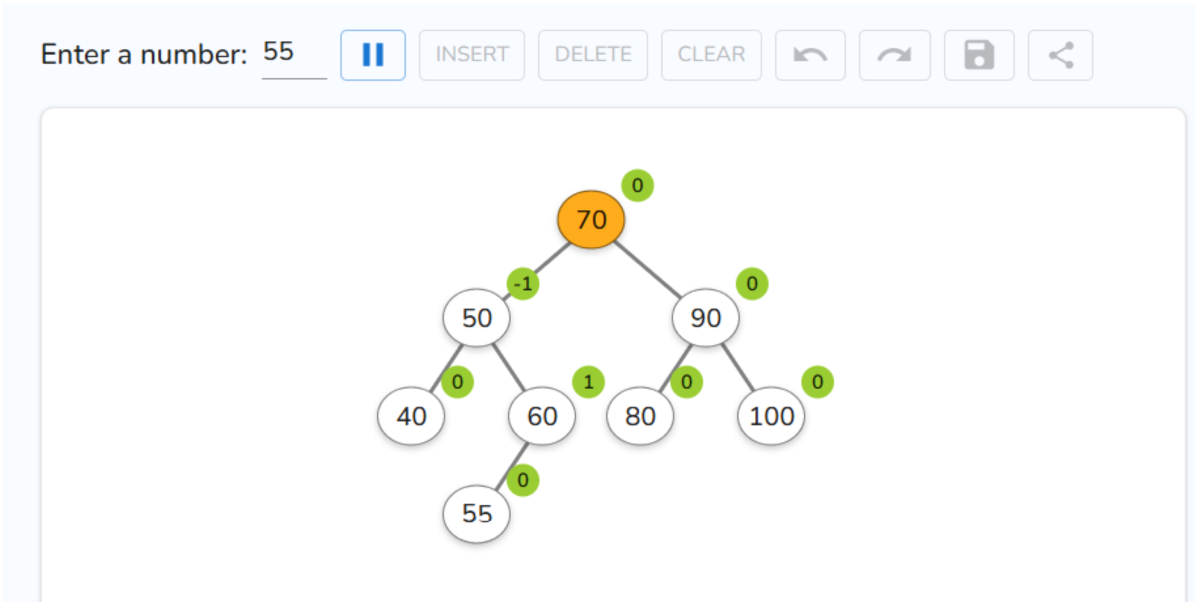

Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
50
30
70
20
40
60
80



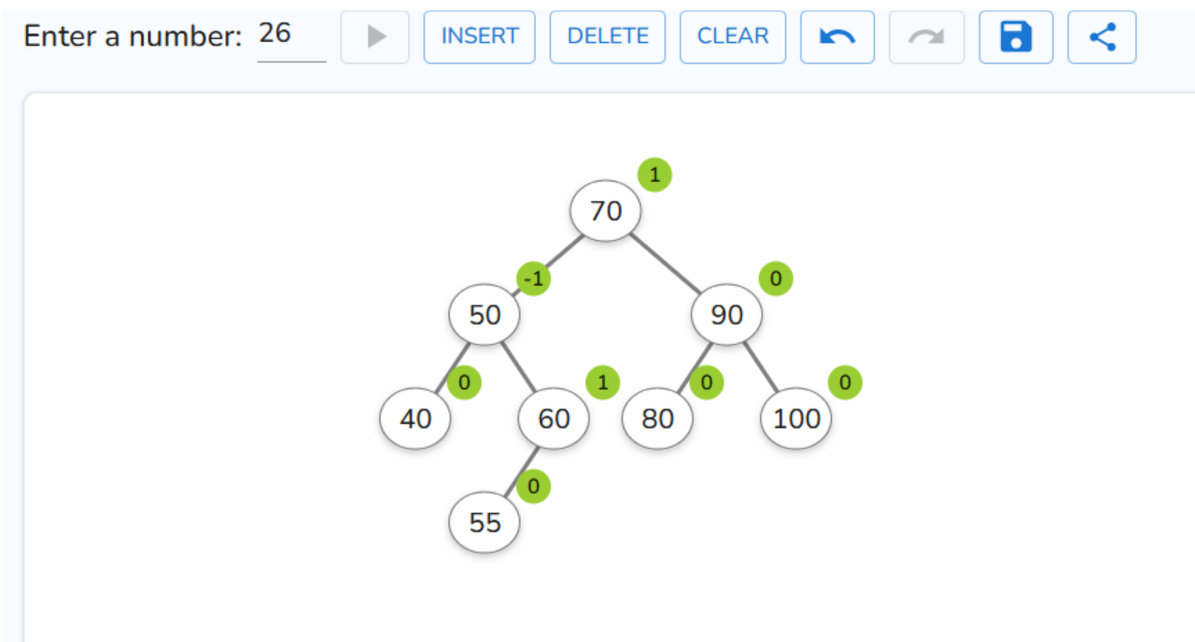
Preorder

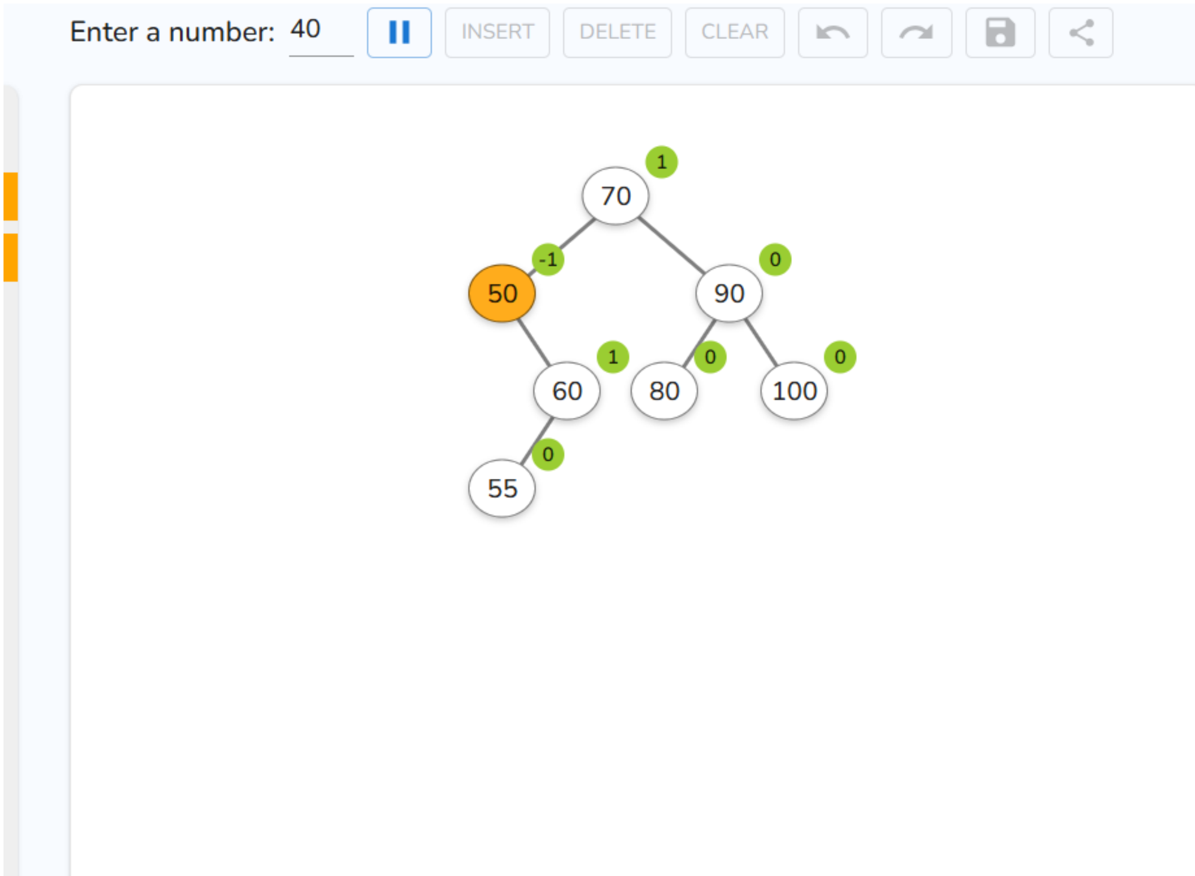


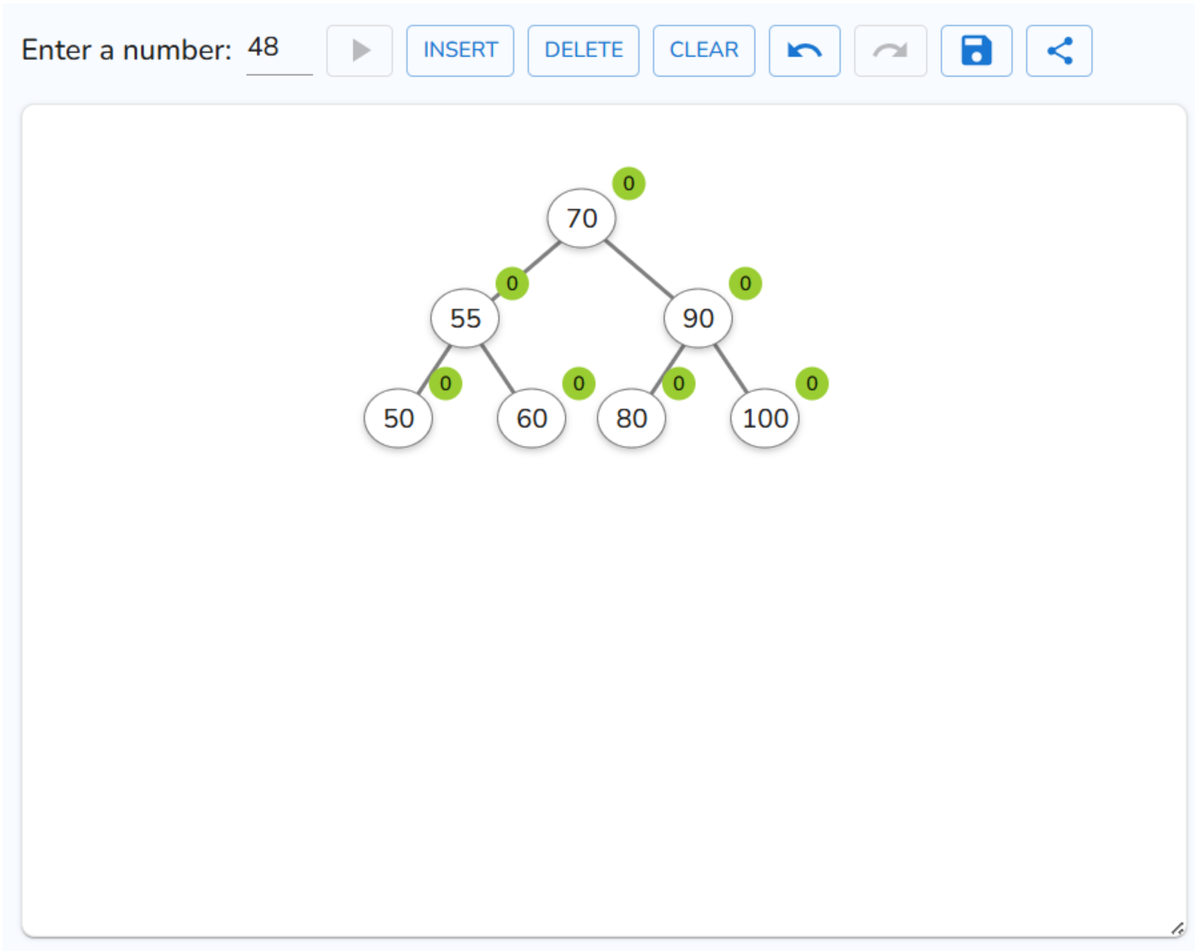
3. AVL Tree


Enter a number: 55
INSERT
DELETE
CLEAR
70
0
-1
50
90
0
40
0
60
1
80
0
100
0
55
0

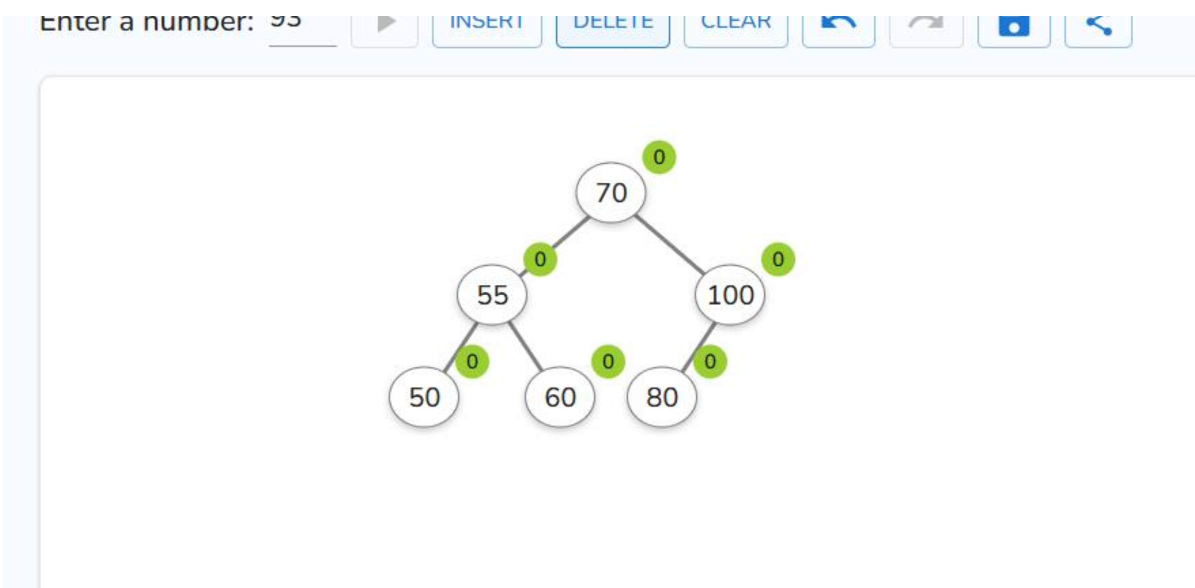
DELETE 40

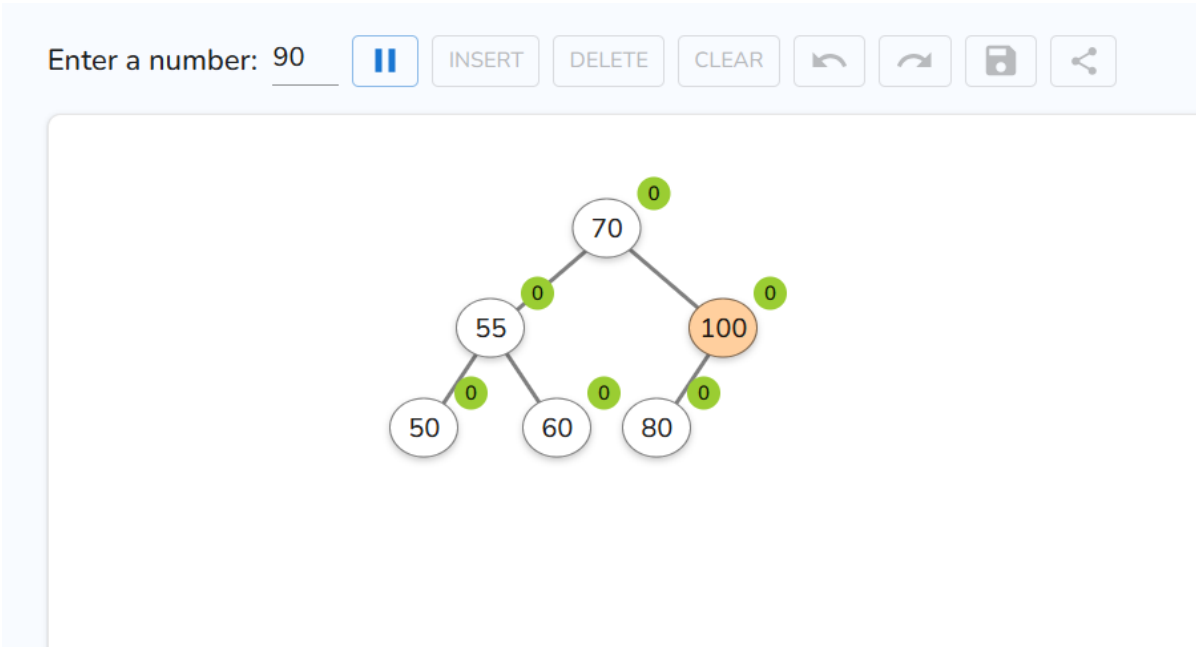

Enter a number: 26
INSERT
DELETE
CLEAR
70
1
-1
50
90
0
40
0
60
1
80
0
100
0
55
0


Enter a number: 40
INSERT
DELETE
CLEAR
70
1
50
-1
90
0
60
1
80
0
100
0
55
0

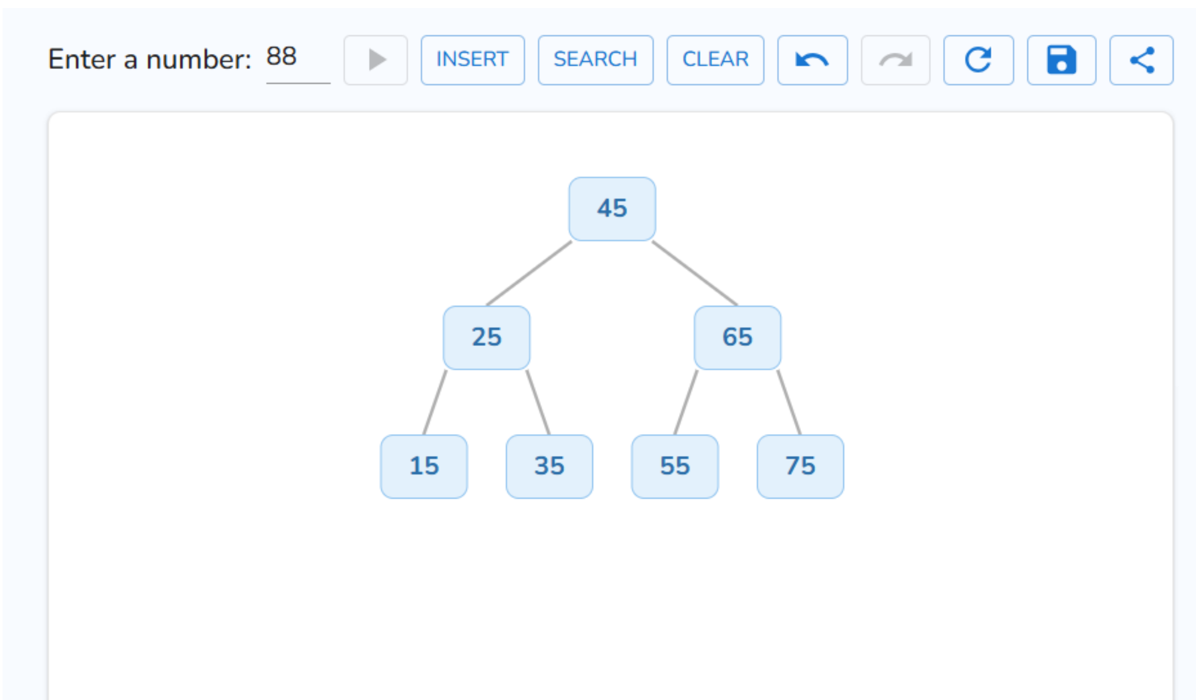

Enter a number: 48
INSERT
DELETE
CLEAR
70
0
55
90
50
60
80
100

DELETE 90


70
0
55
100
50
60
80



4.Construct a B-Tree



5.Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree